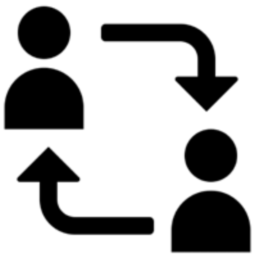
A 2.2.3 Networking Models



A 2.2.3 Compare and contrast networking models.

Networks use two main models: client-server (central server provides services) and peer-to-peer (devices share resources directly). Client-server offers control and scalability, while P2P is resilient and cost-effective. Web browsing uses client-server, while file-sharing can use both. Each model has implications for security and data privacy.

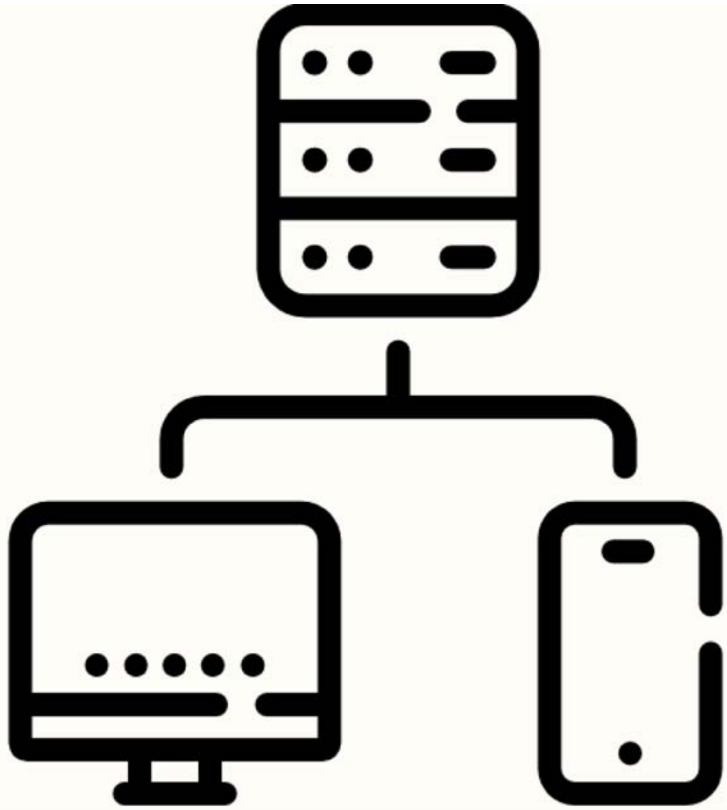
How do client-server and peer-to-peer (P2P) networking models compare and contrast?

When computers connect and communicate, they follow specific models defining their interactions. Two primary models are client-server and peer-to-peer. Understanding their differences is key to grasping how various network applications function.

What are the benefits and drawbacks of the client-server and peer-to-peer models?

Client-Server Model

How it works: A central server provides services (like files, email, or web pages) to multiple clients (like your computer). Clients request services, and the server responds.



Benefits

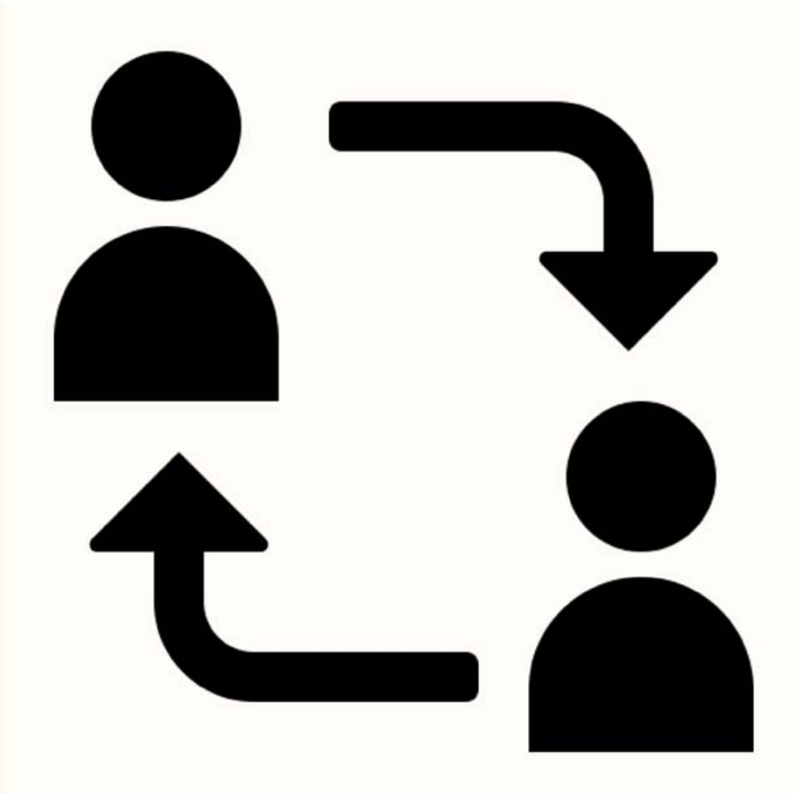
- **Centralised control:** Easier to manage, update, and secure.
- **Scalability**: Can handle many clients and large amounts of data.
- **Data backup**: Easier to back up data stored on a central server.

Drawbacks

- **Single point of failure**: Clients lose access if the server goes down.
- **Cost**: Servers can be expensive to set up and maintain.
- **Potential bottleneck:** The server can become overloaded if too many clients request services simultaneously.

Peer-to-Peer (P2P) Model

How it works: No central server; each computer (peer) acts as a client and a server, sharing resources directly with peers.



Benefits

- **No single point of failure:** More resilient since no central server can fail.
- **Cost-effective:** No need for expensive server hardware.
- **Distributed resources:** Utilises the resources of all peers.

Drawbacks

- **Security concerns:** Can be more vulnerable to malware and unauthorised access.
- **Difficult to manage:** Managing a large P2P network can be complex.
- **Performance can vary:** Depends on the availability and resources of individual peers.

How are these models applied in real-world scenarios, such as web browsing, email, online banking, file sharing, VoIP, and blockchain?

Web browsing

Primarily client-server: Your web browser (the client) sends requests for web pages to a web server. The server then sends the requested page back to your browser. This model efficiently delivers content to many users, as the server can handle multiple requests simultaneously.
Example: When you type www.google.com into your browser, your browser sends a request to Google's web servers. The servers then send back the Google homepage to your browser.

Email

Typically client-server: Your email client (like Outlook or Gmail) connects to a mail server to send and receive emails. The mail server acts as a central hub for storing and managing emails.
Example: When you send an email, your email client sends it to your mail server. The mail server then forwards the email to the recipient's mail server, which stores it until the recipient retrieves it with their email client.

Online banking

Client-server: You access your bank's services (like checking your balance, transferring funds, or paying bills) through a secure connection to their server. The server handles your requests and provides you with the necessary information or performs the requested actions.
Example: When you log in to your online banking account, your computer sends your login credentials to the bank's server. The server verifies your credentials and grants you access to your account information.

File sharing

This can use both of the models.
Client-server: Services like Dropbox, Google Drive, and OneDrive use a client-server model. You upload files to their servers, and then you can access them from any device. This provides centralised storage and backup.
P2P: Torrenting uses a P2P model. When you download a torrent, you're downloading pieces of the file from many different users who are also sharing that file. This distributes the load and can make downloads faster.

VoIP (Voice over IP)

Can be both models:
Client-server: Some VoIP services (like Skype or Zoom) use a client-server model for call setup and management. The server helps establish user connections and may also handle features like call recording or video conferencing.
P2P: Other VoIP services may use a P2P model for direct communication between users once the call is established. This can reduce latency and improve call quality.

Blockchain

P2P: Blockchain technology relies on a P2P network to distribute and validate transactions. Each computer on the network (a node) has a copy of the blockchain, which is a record of all transactions. When a new transaction occurs, it's broadcast to the network, and the nodes work together to verify its validity. This decentralised approach makes blockchain secure and transparent.
Example: In Bitcoin, a cryptocurrency that uses blockchain, transactions are verified and added to the blockchain by a network of computers. This eliminates the need for a central authority like a bank.

Key Questions

What are the key differences in how client-server and P2P networks handle data and resources?

Centralization vs. Distribution: Client-server systems rely on a central server to manage and distribute resources, while P2P systems distribute resources among all peers.
Request-Response vs. Sharing: Client-server uses a request-response model where clients request services from the server, while P2P involves direct sharing between peers.
Control and Management: Client-server offers centralized control, while P2P is more decentralized.

What are each model's advantages and disadvantages regarding security and scalability?

Security:

- Client-server: Easier to implement security measures on a central server, but the server itself can be a target.
- P2P: More difficult to secure due to the distributed nature, making it vulnerable to malware and unauthorized access.

Scalability:

- Client-server: Generally more scalable, as servers can be upgraded to handle increased loads.
- P2P: Scalability can be limited by the resources and bandwidth of individual peers.

Can you give more examples of applications that use each model and why that model is suitable?

Client-server:

- Online games (with a central game server) - for managing game state and player interactions.
- Video streaming services (like Netflix) - for efficient content delivery to a large audience.
- Cloud storage services (like Google Drive) - for centralized data storage and access control.

P2P:

- File-sharing applications (like BitTorrent) - for efficient distribution of large files.
- Cryptocurrency networks (like Bitcoin) - for decentralized transaction validation and security.
- Decentralized social media platforms - for resisting censorship and control.

How do hybrid models combine aspects of both client-server and P2P?

Example: Instant messaging applications (like Skype) might use a central server for user authentication and presence information but use P2P for direct user communication.
Benefits: Can leverage the strengths of both models, such as centralized control and distributed resource sharing.

What are the implications of each model for data privacy and security?

Client-server:

- Centralised data: A significant amount of data is often stored on the server, making it a prime target for attackers. A data breach on the server could compromise the privacy of many users.
- Reliance on the provider: Users rely on the server provider to implement strong security measures and protect their data.
- Security is easier to manage on a central server: Security measures (like firewalls, intrusion detection systems, and access controls) can be implemented and managed more easily on a central server.

P2P:

- Distributed data: Data is distributed among many peers, making it potentially more difficult for attackers to target a single point of failure. However, each peer needs to ensure the security of their own data.
- Shared responsibility: Security becomes a shared responsibility among all peers in the network.
- Vulnerability to malicious peers: Malicious peers can spread malware or compromise the privacy of other network users.

How do these models influence the design and functionality of online services and applications?

Client-server:

- Centralised services: Enables services that require centralized control, authentication, and data management, such as online banking, email, and social media platforms.
- Scalability and reliability: Allows for easier scaling of services to accommodate a large number of users and provides a higher level of reliability due to dedicated server resources.
- User accounts and profiles: This feature facilitates the creation of user accounts and profiles, allowing for personalized experiences and access control.

P2P:

- Decentralised applications: Enables applications that don't rely on a central authority, such as file-sharing, cryptocurrency networks, and some online games.
- Resilience and fault tolerance: This offers greater resilience to server failures, as the network can continue to function even if some peers are unavailable.
- Direct user interaction: Allows for direct interaction and collaboration between users, such as in P2P file-sharing or online gaming.

Quiz

1

Which network model uses a central server to provide services to clients?

A. Client-server

B. Hybrid

C. Mesh

D. Peer-to-peer

2

Which of the following is NOT a benefit of the client-server model?

A. Scalability

B. No single point of failure

C. Easier data backup

D. Centralised control

3

Which network model is more resilient to server failures?

A. Star

B. Peer-to-peer

C. Hybrid

D. Client-server

4

Which of the following is a drawback of the peer-to-peer model?

A. Single point of failure

B. Requires a central server

C. Difficult to manage

D. High cost

5

Which network model is typically used for web browsing?

A. Ring

B. Client-server

C. Peer-to-peer

D. Hybrid

6

Which of the following applications can use BOTH client-server and peer-to-peer models?

A. Online banking

B. Email

C. Web browsing

D. File sharing

7

Which network model is used by blockchain technology?

A. Bus

B. Peer-to-peer

C. Client-server

D. Hybrid

8

Which of the following is a security concern associated with the peer-to-peer model?

A. Vulnerability to malicious peers

B. Reliance on a single server

C. Centralised data storage

D. High cost of security implementation

9

How does the client-server model influence the design of online services?

A. It makes scaling services difficult.

B. It increases the risk of server failures.

C. It promotes decentralisation.

D. It enables centralised authentication and user accounts.

10

What is an implication of the client-server model for data privacy?

A. A data breach on the server could compromise many users' data.

B. Data is distributed, making it difficult to track.

C. Users have full control over their data.

D. There are no privacy concerns with the client-server model.

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